Generalization of Plant Disease Classification Across a Lab-to-Field Domain Shift

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***Abstract*—This study investigates plant disease classification under a lab-to-field domain shift. While models trained on controlled laboratory datasets such as PlantVillage achieve near-perfect performance, their effectiveness degrades significantly on real-world field data such as PlantDoc. To analyze this gap, a unified benchmark dataset was constructed and three approaches were evaluated: a ResNet-18 trained from scratch, staged transfer learning with ResNet-50, and EfficientNet-B0 with CutMix augmentation. Grad-CAM was also used to qualitatively assess model behavior across domains. Results show that all models perform well on in-distribution data but suffer a substantial drop on out-of-distribution samples. Transfer learning with strong augmentation and limited few-shot (FS) adaptation provided the strongest robustness, emphasizing the importance of training strategies that prioritize real-world generalization.**

I. Introduction

In many real-world machine learning applications, labeled datasets are first collected in controlled settings where imaging conditions, backgrounds, and viewpoints are easier to standardize. Deployment, however, rarely occurs under the same conditions. In agriculture, this mismatch is especially evident when models trained on clean leaf images are expected to operate on cluttered field imagery. As a result, strong in-distribution performance does not necessarily translate to reliable real-world use.

This project studies that gap using PlantVillage [1], [2] as a laboratory-style source domain (In-Distribution, or ID) and PlantDoc[1] [6] as a field-style target domain (Out-Of-Distribution, or OOD). To support this study, we constructed a unified benchmark dataset, PlantLab2RealGeneralization, available on Kaggle: PlantLab2RealGeneralization. The central problem is therefore not only classification accuracy, but whether knowledge learned from controlled laboratory imagery can generalize to visually different field data.

II. Related Works

A. Datasets

Several datasets have been developed for plant disease detection, among which PlantVillage and PlantDoc are the most widely used.

1) PlantVillage: The PlantVillage dataset is a widely used benchmark for plant disease classification, containing over 54,000 images across 14 crop species and 38 classes. It includes healthy and diseased leaves collected under controlled conditions with uniform backgrounds and consistent lighting, and is therefore commonly treated as a laboratory-style dataset. [7]

2) PlantDoc: The PlantDoc dataset contains approximately 2,500 images from 13 plant species and around 30 classes. Unlike PlantVillage, these images were collected in real-world field conditions and therefore include variation in lighting, background, occlusion, and image quality. This makes PlantDoc more representative of deployment, but also more challenging due to noise, occasional annotation issues, and class imbalance. [7]

B. Convolutional Neural Networks

Deep learning has become a dominant approach for plant disease classification because CNNs can automatically learn relevant visual features without manual feature engineering. Since deep models often require large datasets and significant computation, many studies use transfer learning from large-scale datasets such as ImageNet.

Among CNN architectures, ResNet models are especially common because their residual connections improve training stability in deeper networks. Variants such as ResNet18 and ResNet50 are therefore widely used as strong baselines for plant disease classification. [8]

A key challenge in recent literature is the domain gap between controlled laboratory images and real-world field imagery. Ahmad *et al.* showed that generalization can improve when field data are incorporated and background effects are reduced. [8] Jeon *et al.* proposed lab-to-field adaptation using background recomposition and unsupervised domain adaptation. [9] Mahapatra *et al.* also studied intra-dataset and cross-dataset training across PlantVillage, PlantDoc, and related datasets, showing improved robustness under dataset shift with an attention-based CNN. [10] Saleem *et al.* presented a transfer-learning framework for plant disease localization and classification. [11] Together, these studies motivate evaluating both laboratory and field-image performance in the present work.

III. Materials and Methods

Three approaches were compared to address the generalization problem. Each method is outlined below, and gradient-

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[1]PlantDoc was the primary field-style domain, other datasets were used as well [3]–[5]

based visualization was used throughout as qualitative feedback for understanding model behavior.

A. Gradient-Based Visualization methods

Vanilla saliency maps [12] are among the earliest and simplest methods for interpreting deep neural networks. They estimate how sensitive a model's prediction is to each input pixel by computing the gradient of the class score with respect to the input image. In practice, larger gradient magnitudes indicate pixels with greater influence on the prediction, although these maps can be noisy and may fail to emphasize the most relevant regions.

Class Activation Mapping (CAM) identifies discriminative image regions using a weighted combination of feature maps from the final convolutional layer, but it requires a specific architecture with global average pooling directly connected to the classifier. Gradient-weighted Class Activation Mapping (Grad-CAM) removes this restriction by using gradient information to estimate feature-map importance, making it applicable to a wider range of convolutional neural networks. [13]

In this work, Grad-CAM is used to visualize the regions most responsible for a model's prediction. Gradients of the target class score are computed with respect to the feature maps in the last convolutional layer, averaged spatially to produce one importance weight per channel, and combined to form a coarse heatmap. After applying ReLU, the heatmap is normalized, resized to the original image size, and overlaid on the input image for both in-distribution and out-of-distribution samples. [14]

B. Building and Training a CNN from Scratch

As a non-transfer-learning baseline, a ResNet-18 classifier was trained from scratch using only the constructed PlantLab2RealGeneralization dataset. ResNet-18 was selected as an established architecture so that the comparison could focus on data and training strategy rather than on a novel network design. No pretrained weights were used, and all network parameters were learned directly from the project dataset.

Because the source data consisted primarily of laboratory-style images, robustness was improved mainly through data augmentation. Resizing, random cropping, and image distortions were applied to reduce background reliance, encourage the model to attend more strongly to the leaf itself, and partially bridge the visual gap between PlantVillage and the more variable field-style images used for out-of-distribution evaluation. Grad-CAM was used as a qualitative diagnostic tool to verify that learned attention patterns were reasonable and not dominated by obvious background shortcuts.

Training used cross-entropy loss with class weighting, the AdamW optimizer, an initial learning rate of 10^{-3}, and a ReduceLROnPlateau scheduler. The model was trained for up to 40 epochs with early stopping based on validation loss, and the best-performing checkpoint was saved for final evaluation. For comparison, a small few-shot (FS) adaptation stage was also evaluated after the initial Test_OOD results were obtained. A held-out portion (roughly 10%) of the Test_OOD dataset was further split into 80% training and 20% validation and used for FS training over 10 epochs. Because this baseline was trained entirely from scratch on a limited and visually specific dataset, it was expected to be more fragile than pretrained alternatives.

C. Transfer Learning with ResNet-50 and Test-Time Augmentation

ResNet-50 is used as the base architecture due to its strong ImageNet performance and favorable balance between accuracy and computational cost [15]. The original 1000-class fully connected layer is replaced with a task-specific classifier of the form

$$\text{FC head: } 2048 \to 512 \to C,$$

where $C = 30$ is the number of leaf classes. More specifically, the 2048-dimensional feature vector produced after global average pooling is passed through a dropout layer, a linear layer $(2048 \to 512)$, batch normalization, and a ReLU activation, followed by a second dropout layer and a final linear layer $(512 \to C)$. Compared with a single linear classifier, this deeper head provides additional regularization and adaptation capacity when transferring from ImageNet to a smaller plant-disease dataset with noticeable domain shift [16], [17].

The ResNet-50 backbone is initialized with ImageNet pretrained weights and is frozen at the start of training. Fine-tuning is then performed in three stages by progressively unfreezing deeper residual blocks.

1) Training Phases: Training is carried out using mixed-precision computation, the AdamW optimizer, label-smoothed cross-entropy loss, and early stopping based on validation loss.

a) Phase 1: Head-Only Training: In the first phase, only the new classification head is trainable, while the pre-trained backbone remains frozen. This allows the newly initialized layers to stabilize before adapting the feature extractor. A learning rate of 10^{-3} is used for the head, and a ReduceLROnPlateau scheduler lowers the learning rate when validation loss stops improving.

b) Phase 2: Unfreezing `layer4`: In the second phase, `layer4` is unfrozen together with the classifier head. This allows high-level semantic features to adapt to plant-specific textures and disease patterns while retaining lower-level ImageNet features. Differential learning rates are used, with a smaller learning rate for `layer4` and a larger one for the classification head. A cosine annealing scheduler with warm restarts is used in this stage.

c) Phase 3: Unfreezing `layer3` and `layer4`: In the final phase, both `layer3` and `layer4` are unfrozen along with the head, enabling broader task-specific adaptation while keeping the earliest layers fixed. Learning rates are reduced further to support stable fine-tuning.

2) Data Augmentation and Regularization: To reduce overfitting and improve robustness to field conditions, strong augmentation is applied during training. These transformations include random resized cropping, horizontal and vertical

flips, rotation, color jitter, grayscale conversion, Gaussian blur, perspective distortion, and random erasing. Together, these augmentations simulate variation in framing, viewpoint, illumination, blur, and partial occlusion.

All images are normalized using channel-wise mean and standard deviation estimated from the training split. During evaluation, a deterministic transform consisting of image resizing and normalization is applied. In addition, test-time augmentation (TTA) is evaluated on the OOD test set by generating multiple stochastic views of each image and averaging the predicted class probabilities across 7 augmented samples.

To further address class imbalance, the training loader uses a weighted random sampler so that underrepresented classes are seen more frequently during optimization.

3) Metrics: Finally, as discussed in section IV, accuracies were evaluated on both the in-distribution (lab data) and out-of-distribution (field data) splits, with and without TTA, along with confusion matrices for class-level analysis.

D. CutMix Augmentation with EfficientNet Model

An ImageNet-pretrained EfficientNet-B0 was fine-tuned for plant disease classification due to its strong accuracy–efficiency trade-off and good transfer learning performance. The final classifier was replaced with a task-specific linear layer matching the 30 plant disease classes. Training used dataset-specific normalization together with moderate augmentation, including random resized cropping, horizontal flipping, rotation, and color jitter.

To further improve robustness, CutMix was applied to 50% of training batches. This augmentation encourages the model to rely on spatially distributed disease features rather than background correlations by combining image regions from two samples and mixing their labels proportionally. Class imbalance in the PlantVillage training split was addressed using inverse-frequency class weighting in the cross-entropy loss.

Optimization was performed using AdamW with a learning rate of 3×10^{-4}. After the main training stage, a short FS adaptation phase was performed using a small set of PlantDoc-style images. During this stage, the EfficientNet feature extractor was frozen and only the classifier was updated using Adam with a reduced learning rate of 1×10^{-4} for 5 epochs. This allowed limited adaptation to the target domain while reducing the risk of catastrophic forgetting.

Finally, Grad-CAM was used to qualitatively compare model attention on in-distribution PlantVillage samples and out-of-distribution PlantDoc samples, helping assess whether the model focused on meaningful disease regions under both conditions.

IV. Results and Discussion

All three model families achieved very high accuracy on the laboratory-style PlantVillage domain, but all suffered a clear drop when evaluated on field imagery from PlantDoc. This confirms that the main challenge in this project was not in-distribution classification, but cross-domain generalization from clean laboratory images to more variable real-world images.

TABLE I: Summary of model performance on Test_ID and Test_OOD.

Model	Test_ID	Test_OOD	Adapted by FS
ResNet-18 (scratch)	99.29	13.14	71.51 / 17.10
ResNet-50 (transfer)	99.49	24.73	95.58 / 45.58
EfficientNet-B0 + CutMix	99.76	23.88	98.08 / 28.26

The ResNet-18 model trained from scratch achieved 99.29% accuracy on Test_ID, with near-perfect class separation in the confusion matrix shown in Figure 1. However, this strong in-distribution performance did not transfer well to field imagery, where Test_OOD accuracy dropped to 13.14%. FS fine-tuning improved Test_OOD only marginally to 17.10%, while reducing Test_ID to 71.51%, indicating severe catastrophic forgetting. These results show that, although a scratch-trained model can fit the source domain extremely well, it remains too fragile for reliable adaptation under strong domain shift.

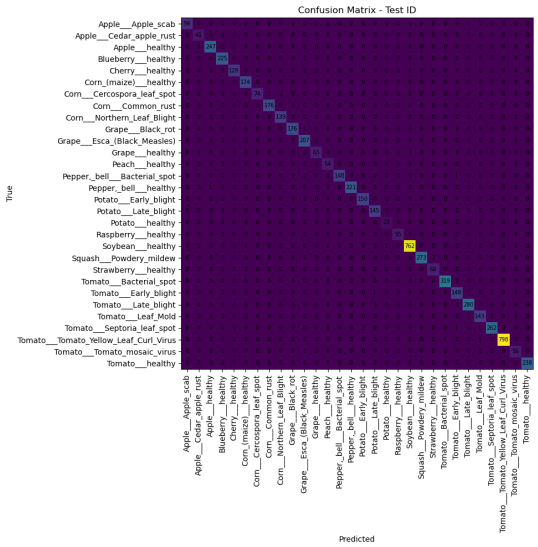

Fig. 1: Confusion matrix for the ResNet-18 model trained from scratch and evaluated on the in-distribution PlantVillage test set.

As described in section III-A, Grad-CAM was used to assess whether the model was learning meaningful image features rather than relying on background noise or other spurious cues. As shown in Figure 2, the ResNet-18 model generally focused on relevant disease regions, leaf texture, and leaf structure, suggesting that the model was identifying useful visual features rather than making arbitrary predictions.

A stronger result was obtained using the transfer-learned ResNet-50 model. As summarized in Table I, this model achieved 99.49% Top-1 accuracy on Test_ID and 24.73% on Test_OOD, substantially outperforming the scratch baseline under domain shift. Applying test-time augmentation further increased Test_OOD accuracy to 26.87%, while Top-5 ac-

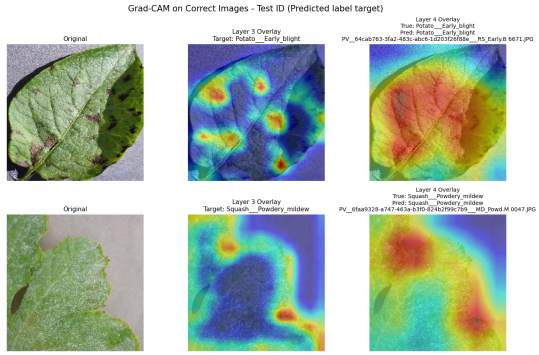

Fig. 2: Grad-CAM examples for the scratch-trained ResNet-18 model on Test_ID samples, showing the original image (left), Grad-CAM from layer 3 (center), and Grad-CAM from layer 4 (right).

curacy improved from 55.81% to 58.68%. The staged fine-tuning strategy also produced stable optimization behavior, as shown in Figure 3. The confusion matrices in Figures 4 and 5 show that classification remained nearly perfect on PlantVillage but degraded considerably on field images. Even so, the transfer-learned model was clearly more robust than the scratch-trained ResNet-18, indicating that ImageNet pretraining provided more transferable visual features. In the final FS stage, OOD accuracy increased further to 45.58%, which was the best cross-domain result obtained in this project.

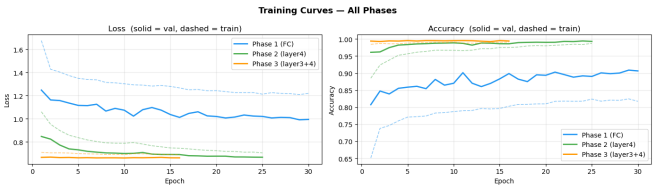

Fig. 3: Training and validation accuracy across the three ResNet-50 fine-tuning phases.

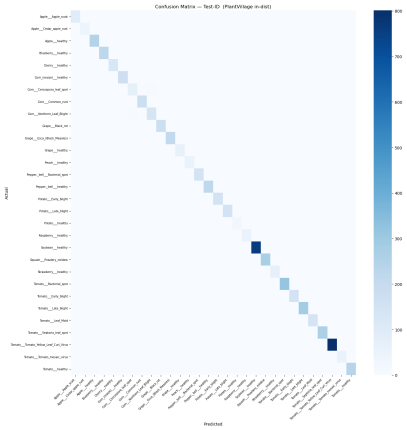

Fig. 4: Confusion matrix for the transfer-learned ResNet-50 model on the in-distribution PlantVillage test set.

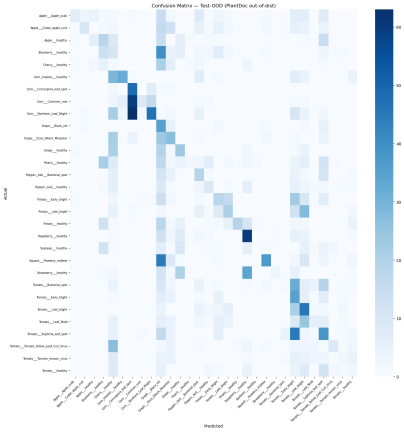

Fig. 5: Confusion matrix for the transfer-learned ResNet-50 model on the out-of-distribution PlantDoc/field test set.

Finally, the EfficientNet-B0 model with CutMix was evaluated as an alternative transfer-learning strategy. Before FS adaptation, it achieved 99.76% accuracy on PlantVillage Test_ID and 23.88% on PlantDoc Test_OOD. After FS fine-tuning, Test_ID decreased only modestly to 98.08%, while Test_OOD improved to 28.26%, corresponding to a gain of 4.38 percentage points. Compared with the scratch-trained ResNet-18, this model was far more stable under adaptation, since OOD performance improved without catastrophic forgetting. However, its final OOD performance still remained below that of the fully adapted ResNet-50 pipeline. Overall, these results indicate that transfer learning was essential for this task, and that the staged ResNet-50 approach provided the best balance between high in-distribution accuracy, improved field generalization, and successful FS adaptation.

V. Conclusions

This study examined the generalization gap between laboratory-style and field-style plant disease imagery using a unified benchmark derived from PlantVillage and PlantDoc datasets. Although all evaluated models achieved near-perfect accuracy on in-distribution laboratory data, performance degraded substantially under field conditions, highlighting the severity of domain shift in agricultural vision tasks.

Models trained from scratch, despite achieving in-distribution accuracies above 98%, exhibited very poor out-of-distribution performance and were highly susceptible to catastrophic forgetting during few-shot adaptation. Transfer learning approaches demonstrated superior robustness compared to models trained from scratch. EfficientNet-B0 with CutMix showed stronger stability during adaptation, while the staged ResNet-50 pipeline achieved the best overall out-of-distribution performance after FS fine-tuning. These results suggest that combining pretrained models, strong augmentation, and lightweight adaptation is an effective strategy for improving real-world reliability in plant disease recognition systems.

REFERENCES

[1] D. P. Hughes and M. Salathé, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," *arXiv preprint arXiv:1511.08060*, 2015.

[2] A. Ali and collaborators, "Plantvillage dataset." https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset, 2019. Kaggle dataset mirror, accessed 2026-04-07.

[3] M. Alessandrini, R. C. F. Rivera, L. Falaschetti, D. Pau, V. Tomaselli, and C. Turchetti, "Esca-dataset," 2021.

[4] R. Acharya, "Corn leaf infection dataset." https://www.kaggle.com/datasets/qramkrishna/corn-leaf-infection-dataset, 2021. Kaggle dataset, accessed 2026-04-07.

[5] N. H. Shabrina, S. Indarti, R. Maharani, D. A. Kristiyanti, I. Irmawati, N. Prastomo, and T. A. M, "Potato leaf disease dataset in uncontrolled environment," 2023.

[6] D. Singh, N. Jain, P. Jain, P. Kayal, S. Kumawat, and N. Batra, "Plantdoc: A dataset for visual plant disease detection," in *Proceedings of the 7th ACM IKDD CoDS and 25th COMAD*, CoDS COMAD 2020, (New York, NY, USA), pp. 249–253, Association for Computing Machinery, 2020.

[7] E. Moupojou, A. Tagne, F. Retraint, A. Tadonkemwa, D. Wilfried, H. Tapamo, and M. Nkenlifack, "Fieldplant: A dataset of field plant images for plant disease detection and classification with deep learning," *IEEE Access*, 2023.

[8] A. Ahmad, A. El Gamal, and D. Saraswat, "Toward generalization of deep learning-based plant disease identification under controlled and field conditions," *IEEE Access*, vol. 11, pp. 9042–9057, 2023.

[9] W. Jeon, T. Kim, S. Choi, K. Yang, S.-Y. Kim, and M. Song, "Bridging the lab-to-field gap in plant disease diagnosis through unsupervised domain adaptation enhanced by background recomposition," *Ecological Informatics*, vol. 93, p. 103579, 2026.

[10] P. Mahapatra, M. Panda, S. K. Dash, and U. K. Sahu, "Advancing plant disease classification using an attention-based cnn for intra-dataset and cross-dataset training," *Scientific Reports*, vol. 16, p. 10925, 2026.

[11] M. H. Saleem, S. Khanchi, J. Potgieter, and K. M. Arif, "Image-based plant disease identification by deep learning meta-architectures," *Plants*, vol. 9, no. 11, p. 1451, 2020.

[12] C. Molnar, *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. Christoph Molnar, 2025.

[13] H. Wang, Z. Wang, M. Du, F. Yang, Z. Zhang, and X. Hu, "Score-cam: Score-weighted visual explanations for convolutional neural networks," *arXiv preprint arXiv:1910.01279*, 2019.

[14] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," *arXiv preprint arXiv:1610.02391*, 2019.

[15] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016.

[16] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, 2016.

[17] K. P. Ferentinos, "Deep learning models for plant disease detection and diagnosis," *Computers and Electronics in Agriculture*, 2018.